

LLOYDMINSTER, AB, Canada

WMO: 718710

Lat: 53.3120N

Lon: 110.0744W

Elev: 669

StdP: 93.55

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-29.2	-36.7	0.1	-31.8	-33.7	0.2	-29.1	12.4	-5.1	10.6	-5.8	3.9	280	0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	28.1	17.4	26.3	16.8	24.6	16.0	19.3	25.6	18.2	24.0	17.2	22.7	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.2	21.8	16.0	12.3	21.1	15.0	11.5	19.8	58.1	25.8	54.3	24.3	50.9	22.7	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.5	9.4	8.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-35.0	31.5	3.2	2.3	-37.3	33.2	-39.2	34.6	-41.0	35.9	-43.4	37.6
			-35.1	21.6	3.2	1.4	-37.4	22.6	-39.3	23.4	-41.1	24.2	-43.4	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-13.9	-12.4	-6.3	3.0	10.2	14.7	17.2	16.0	11.0	3.5	-5.8	-12.3	(6)
(7)		DBStd	12.81	8.97	7.58	7.60	5.59	4.59	3.27	2.94	3.40	4.54	4.92	6.73	8.04	(7)
(8)		HDD10.0	3566	742	627	507	216	54	4	0	2	41	208	473	693	(8)
(9)		HDD18.3	5952	1000	860	765	459	252	117	57	86	221	461	723	951	(9)
(10)		CDD10.0	701	0	0	0	6	62	144	223	189	72	6	0	0	(10)
(11)		CDD18.3	46	0	0	0	0	2	6	21	15	2	0	0	0	(11)
(12)		CDH23.3	663	0	0	0	2	52	98	238	211	62	1	0	0	(12)
(13)		CDH26.7	134	0	0	0	0	8	17	48	49	12	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.4	4.4	4.6	4.7	4.6	4.2	4.0	3.9	4.3	4.5	4.5	4.4	(14)
(15)	Precipitation	PrecAvg	409	18	12	16	29	45	74	79	58	34	17	15	19	(15)
(16)		PrecMax	593	39	31	41	83	123	149	233	170	95	42	49	33	(16)
(17)		PrecMin	222	4	2	4	3	6	13	14	4	3	2	1	5	(17)
(18)		PrecStd	90	10	7	8	22	30	31	46	31	25	11	12	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	4.3	10.5	22.1	27.6	28.6	30.6	31.1	28.4	20.7	10.8	5.7	(19)
(20)			MCWB	1.7	1.5	5.2	11.0	14.2	17.2	19.5	18.6	16.0	12.2	6.1	2.2	(20)
(21)		2%	DB	2.9	2.0	6.7	18.4	24.6	26.0	27.7	27.7	24.9	16.3	7.0	3.1	(21)
(22)			MCWB	0.8	0.0	2.9	9.2	13.0	16.0	18.8	17.4	14.6	9.6	3.6	0.9	(22)
(23)		5%	DB	0.9	0.1	4.5	15.2	21.9	23.9	26.0	25.7	22.2	13.6	4.5	1.0	(23)
(24)			MCWB	-0.7	-1.5	1.8	7.3	11.8	15.1	17.9	16.9	13.5	7.8	1.9	-0.8	(24)
(25)		10%	DB	-1.7	-1.8	2.8	12.2	19.4	22.0	24.1	23.6	19.4	11.0	2.3	-1.5	(25)
(26)			MCWB	-3.0	-3.2	0.7	6.0	10.7	14.1	17.1	16.2	12.3	6.4	0.4	-2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.0	5.7	11.8	15.6	19.1	21.5	20.2	16.9	12.7	6.4	2.6	(27)
(28)			MCD B	4.2	3.9	9.6	20.8	24.3	25.2	27.4	27.1	25.4	20.0	10.1	5.1	(28)
(29)		2%	WB	0.9	0.1	3.3	9.6	13.9	17.3	19.8	18.8	15.3	10.2	4.0	1.0	(29)
(30)			MCD B	2.8	1.7	6.2	17.6	22.1	23.4	26.3	25.1	23.2	15.0	6.8	3.0	(30)
(31)		5%	WB	-0.7	-1.4	2.0	7.9	12.6	16.2	18.8	17.7	14.1	8.3	2.1	-0.8	(31)
(32)			MCD B	0.9	0.1	4.1	14.5	20.4	21.7	24.6	23.9	21.3	12.8	4.1	1.0	(32)
(33)		10%	WB	-3.0	-3.1	0.8	6.2	11.4	15.1	17.7	16.7	12.7	6.8	0.5	-2.8	(33)
(34)			MCD B	-1.7	-1.8	2.7	11.6	18.3	20.2	22.8	22.5	18.5	10.7	2.2	-1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.8	8.6	10.8	12.5	11.0	11.3	11.8	11.9	10.0	7.6	7.8	(35)
(36)			MCD BR	9.3	10.4	9.9	16.0	16.2	14.5	14.2	15.5	16.9	14.5	10.1	10.0	(36)
(37)			MCWBR	7.8	9.1	7.1	8.9	7.7	6.9	7.0	7.2	8.3	8.7	7.2	8.3	(37)
(38)		5% WB	MCD BR	9.5	10.2	9.5	15.4	14.4	12.3	12.9	13.6	15.5	13.7	9.3	10.0	(38)
(39)			MCWBR	8.2	9.0	7.0	8.9	7.2	6.6	6.9	7.0	8.0	8.5	7.0	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.287	0.283	0.324	0.346	0.363	0.361	0.370	0.308	0.273	0.247	0.215	(40)	
(41)		taud	2.338	2.206	2.382	2.382	2.368	2.346	2.378	2.332	2.489	2.555	2.453	2.366	(41)	
(42)		Ebn at Noon	772	811	913	910	902	886	881	850	876	848	764	748	(42)	
(43)		Edh at Noon	55	88	92	106	113	117	112	110	83	62	49	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.99	2.06	3.56	4.71	5.81	5.92	6.14	5.00	3.53	2.05	0.98	0.73	(44)	
(45)		RadStd	0.10	0.14	0.27	0.42	0.43	0.43	0.35	0.39	0.39	0.31	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+2.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page